

MUHAMMAD AHMED

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EDUCATION

Korea Advanced Institute of Science and Technology

BS Civil and Environmental Engineering, and Computer Science(Minor in Business and Technology Management)

Daejeon, South Korea

Sep 2021 - Present

FFC Grammar School and College

O/A levels Engineering and Business

Mirpur Mathelo, Pakistan

Aug 2005 - Aug 2021

WORK EXPERIENCE

KAIST MetaEarth Research Center

Research Intern

Daejeon, South Korea

May 2024 - Present

- Conducting research on lightweight AI-based global weather forecasting; helped develop **KAI- α** , an ultralight CNN model achieving competitive skill with $8\times$ fewer parameters than state-of-the-art models such as GraphCast and FourCastNet. Built scalable end-to-end training and inference pipelines from scratch, including modular evaluation frameworks enabling efficient benchmarking across diverse model architectures.
- Studying physics-informed statistical downscaling; contributed to **PcMAN-DS**, a multi-scale, constraint-aware model for surface climate variables, enabling reliable resolution enhancement from 2° to 0.25° .
- Vice Lead of the Data-Driven Team – Spring 2025

KAIST AIxMobility Lab

Undergraduate Assistant

Daejeon, South Korea

March 2023 - February 2024

- Generated and validated high-resolution orthophotos to support urban mapping and large-scale dataset development, improving data quality and spatial consistency (Haechan 2024).
- Developed computer vision methods for automated road condition monitoring, enabling scalable detection and analysis of roadway features.
- Contributed to the design and implementation of an AI-driven traffic simulation system, supporting data-informed analysis of urban mobility and traffic dynamics.

PROJECTS

Development and Evaluation of Graph Neural Network Based Climate Downscaling Methods

Undergraduate Research Program Spring, 2025

- Implemented graph-based climate downscaling models for ERA5 over South Asia, building a multi-scale graph reasoning setup and a spectral-structural loss to keep high-frequency features accurate.
- Reached 60–85% RMSE reduction across winds, temperature, humidity, and radiative fluxes, and about 84% for surface pressure.
- Kept spectral errors within $\pm 0.2\%$ across 90% of wavenumbers.

Spatio-Temporal Drought Prediction Using Graph Neural Networks

[GitHub]

Graph Machine Learning and Mining Spring 2025

- Designed a two-stage spatio-temporal architecture: (1) GRU-based temporal encoder over $T=6$ steps followed by shared GraphSAGE layers for next-state climate prediction, and (2) GraphSAGE-based classifier for drought severity classification.
- Modeled temporal dynamics of key climate variables and generated structured predictions for downstream drought classification (classes 0–4).
- Improved classification accuracy from 40% to 70% over bilinear and CNN baselines, highlighting the effectiveness of spatio-temporal graph modeling.

Accessibility and Environmental Impact Analysis of the US Northeast Megaregion

Urban and Regional Sustainability Spring 2024

- Analyzed public-transit accessibility and interstate travel emissions across major Northeast cities using spatial datasets and network mapping.
- Assessed 30-minute job accessibility and compared modal travel times with transport CO₂ emissions.
- Found stronger transit connectivity in New York and D.C., with higher car dependence and emissions in Baltimore and Philadelphia.
- Evaluated policy scenarios, estimating up to 29.5 kg CO₂ savings per trip when shifting from car travel to rail.

Detecting Fraud in Customs Declarations in Korea
[GitHub]

Introduction to Data Science Spring, 2023

- Applied statistical methods to analyze historical customs records and identify fraudulent declarations.
- Reached 88% classification accuracy using KNNs on a real-world dataset.

SKILLS

Programming:	Python, C++, Java, Rust, MATLAB
Machine Learning:	PyTorch, TensorFlow, Super Resolution, Spatio-temporal Modeling
Data Processing:	NumPy, Pandas, SQL, Xarray, Cartopy, Xesmf, Model Evaluation
Geospatial and Vision:	QGIS, MetaShape, OpenCV
Systems:	ROS, Training & Inference Pipelines
Languages:	English (Fluent), Urdu (Native), Punjabi (Native), Korean (Intermediate)